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Question	Answer	Marks
1	**Defining the problem** θ is the independent variable and v is the dependent variable, or vary θ and measure v.	**1**
	Keep d constant	**1**
	Methods of data collection Labelled diagram of workable experiment including: • sheet supported by stand / jack • light gate positioned at X • support, light gate and X labelled.	**1**
	Light gate connected to timer / datalogger.	**1**
	Measure length (L) (of card) interrupted by beam for single light gate.	**1**
	Method to measure θ, e.g. use protractor or Method to determine θ, e.g. use a rule(r) to measure two appropriate distances to use in a trigonometrical ratio	**1**
	Method of Analysis Plots a graph of v^2 on y-axis and sin θ on x-axis. **Allow** other valid graphs, e.g. sin θ against v^2 Do not accept log graphs.	**1**
	$p = \frac{\text{gradient}}{2d}$ for v^2 against sin θ or $p = \frac{1}{2d \times \text{gradient}}$ for sin θ against v^2	**1**

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Question	Answer	Marks
1	$q = -\dfrac{m \times y - \text{intercept}}{2Bd}$ for v^2 against sin θ or $q = \dfrac{mp \times y - \text{intercept}}{B} = \dfrac{m \times y - \text{intercept}}{2dB \times \text{gradient}}$ for sin θ against v^2	**1**
	Additional detail including safety considerations Any six from:	**6**
	D1 Method to <u>stop</u> the trolley once the trolley passes X, e.g. place a block / stop on the bench near the end of the sheet Ignore trolley falls	
	D2 Keep *B* <u>and</u> *m* constant	
	D3 Use a rule(r) to measure *d*	
	D4 Method to keep *d* constant, e.g. mark distance *d* on the sheet or the starting position of the trolley on the sheet	
	D5 Method to measure mass of trolley (and magnet), e.g. use balance or use newton meter to measure weight and divide by *g* <u>and</u> Measure *B* using a (calibrated) Hall probe	
	D6 Additional detail on use of Hall probe, e.g. adjust probe until <u>maximum</u> value or measure *B* using Hall probe first in one direction, then in the opposite direction and average	
	D7 Determine *v* (the velocity at X) from *L* / *t* (for a single light gate)	
	D8 Additional detail on measuring θ, e.g. protractor drawn in correct position on diagram, or additional detail on determining θ, e.g. relationship between measured lengths and θ	

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Question	Answer	Marks
1	D9 Relationship valid <u>if</u> a straight line is produced (not passing through the origin)	
	D10 Repeat experiment for each θ <u>and</u> average v.	

Question	Answer	Marks
2(a)	Gradient = $\frac{1}{2\pi fC}$	**1**
2(b)	<table><tr><th>$\frac{1}{R}$ / $10^{-3}\,\Omega^{-1}$</th><th>tan θ</th></tr><tr><td>83 or 83.3</td><td>6.17 or 6.174</td></tr><tr><td>63 or 62.5</td><td>4.51 or 4.511</td></tr><tr><td>45 or 45.5</td><td>3.27 or 3.271</td></tr><tr><td>30 or 30.3</td><td>2.16 or 2.164</td></tr><tr><td>26 or 25.6</td><td>1.86 or 1.857</td></tr><tr><td>23 or 23.3</td><td>1.68 or 1.684</td></tr></table>	**1**
	Absolute uncertainties in $\frac{1}{R}$ from $\pm$ 4 to $\pm$ 1	**1**